

Brian W. Smith

PRINCIPAL SITE RELIABILITY ENGINEER · FEDRAMP HIGH · CLOUD-NATIVE INFRASTRUCTURE

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25+ YEARS IN PRODUCTION	100+ FEDRAMP-HIGH OPENSIFT CLUSTERS OPERATED	FedRAMP HIGH & MODERATE ATO	700+ NIST 800-53 CONTROLS EVALUATED
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SUMMARY

Principal SRE at Red Hat with 25+ years of enterprise infrastructure experience, specializing in FedRAMP High and Moderate compliance, cloud-native platform engineering, and security automation at scale. Deep expertise in OpenShift/Kubernetes, AWS GovCloud, Terraform IaC, and NIST 800-53 Rev 5 control implementation. Proven track record of building greenfield platforms, leading cross-functional teams, and delivering compliance-driven engineering across regulated federal environments. Currently leading FIPS cryptographic validation, Secret Management platform engineering, FedRAMP Moderate remediation, and AI-powered compliance automation across the ROSA Regional Platform.

TECHNICAL SKILLS

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|--------------------------------|--------------------------------------|--------------------------------|
| ▸ OpenShift / Kubernetes (CKA) | ▸ FedRAMP High & Moderate ATO | ▸ NIST 800-53 Rev 5 |
| ▸ AWS GovCloud / PrivateLink | ▸ Terraform / Infrastructure as Code | ▸ Splunk / Dynatrace / Grafana |
| ▸ Python / Bash / Go | ▸ CI/CD (Tekton, Jenkins, GitLab) | ▸ Keycloak / IAM / SSO |
| ▸ Claude AI / MCP / Agent SDK | ▸ GitOps / Helm / Kustomize | ▸ Incident Response / CONMON |

EXPERIENCE

Principal SRE (January 2026 – Present) · Principal SRE Team Lead (January 2025 – January 2026)

- Drove FedRAMP Moderate compliance remediation across the ROSA Regional Platform, implementing 13+ NIST 800-53 Rev 5 controls (AU, AC, SC, SI, RA, IA, CA, CP) and performing control-by-control mapping of HCP Terraform modules against NIST 800-53 Rev 5; deployed a unified Thanos, Prometheus, Alertmanager, and Grafana observability suite via parameterized IaC modules enabling single-variable regional deployments.
- Designed and deployed a Secret Management framework using AWS Secrets Manager and Vault sidecar injection — enforcing least-privilege IAM policies per service and eliminating plaintext secrets across all repositories; built a Python/DynamoDB Sandbox account leasing system that provisions scoped IAM roles and SCPs and auto-reclaims environments, cutting setup time from hours to minutes.
- Architected a 3-tier AWS CodePipeline GitOps hierarchy — a meta-pipeline provisioner that dynamically creates, updates, and destroys per-cluster pipelines from Git commits — paired with ArgoCD ApplicationSet Matrix Generators for zero-config multi-cluster chart delivery; implemented hash-pinned sector-based progressive deployment (integration → staging → production) from a single `config_revision` change.
- Engineered zero-trust cross-cluster messaging via AWS IoT Core MQTT (eliminating VPC peering and Transit Gateway coupling), ECS Fargate bootstrap-then-handoff for fully private EKS cluster initialization, and a 17-module reusable Terraform library; built ephemeral CI environments (full provision → e2e → teardown via OpenShift CI Prow) with k6 load testing and automated S3-baseline regression detection.
- Led FedRAMP COMMON remediation deploying compliance-monkey across Management Clusters; integrated backplane-cli/backplane-api with FedRAMP-compliant auth protocols resolving SRE emergency access blockers; architected AWS PrivateLink for ROSA HCP across all GovCloud regions with cross-account VPC Peering; delivered 10+ HCP technical training sessions.
- Built a fully automated FedRAMP Compliance Agent (Claude Sonnet sub-agent) evaluating ~700 Moderate and ~100 High-only NIST 800-53 Rev 5 controls across ROSA repositories — 7-step workflow (clone → classify → analyze → report → PDF → Drive → Jira/PR), 12 artifacts per run, Trivy CVE integration to auto-downgrade Met controls on CRITICAL/HIGH findings.
- Orchestrated a cross-vendor FIPS validation effort for OCP on EKS — confirming the Go binary dlopen dispatch path to host kernel flags (`/proc/sys/crypto/fips_enabled`) using negative-control testing to distinguish infrastructure-level enforcement from binary-level compliance; collaborated with Red Hat cryptographic maintainers on hardcoded KAT tests for NIST SP 800-131A AES/RSA/ECDSA compliance.
- Deployed Dynatrace Managed Service across FedRAMP environments with OneAgent log forwarding; engineered a Keycloak → Splunk log pipeline for centralized audit trails and real-time authentication security monitoring; authored SCR architecture diagrams and ADRs for External DNS Operator and HCP authorization.

Senior SRE Functional Lead (December 2023 – January 2025) · Senior SRE (November 2021 – December 2023)

- Led a team of 9 SREs; defined and drove agile epics aligned with business priorities — CVE mitigation, annual FedRAMP assessments, Significant Change Requests, and operational toil reduction.
- Deployed Red Hat build of Keycloak (RHBK) to production, authored six comprehensive FedRAMP Boundary Architecture Diagrams, and engineered real-time synthetic availability monitoring for the identity platform.
- Automated AWS Config, GuardDuty, and SecurityHub via Terraform, establishing a hardened FedRAMP security baseline; orchestrated platform-wide migration from PagerDuty to GoAlert, reducing incident response latency.
- Spearheaded production deployment of GoAlert as primary FedRAMP on-call platform (PoC → VPCE network architecture); authored the SSP Access Control validation epic for the ROSA Regional Platform.
- Built greenfield FedRAMP High GovCloud platform on AWS from scratch; led IaC initiative establishing Terraform as the provisioning foundation; delivered ATO workstreams including annual assessment support.
- Championed Transit Gateway hub-and-spoke topology for GovCloud, VPC Flow Logs global rollout, and WAF compliance audit across all high-compliance regions.

OneStream Software

February 2021 – November 2021

FIRST DEVOPS ENGINEER

- Migrated 70+ repositories from TFS to Azure DevOps; implemented SonarQube SAST scanning, GitFlow branch strategy, and Azure Pipelines CI/CD as the organization's founding DevOps engineer.

Immuta

August 2020 – February 2021

CLOUD ENGINEER

- Worked with customers to install and configure Immuta's data security platform across Databricks, Athena, Snowflake, and Redshift; leveraged deep Kubernetes expertise and broad infrastructure background to rapidly onboard across data warehouse technologies.

Fiserv

March 2017 – August 2020

SENIOR SYSTEM ENGINEER LEAD – CLOUD PLATFORM

- Designed and built 16 Pivotal Cloud Foundry foundations supporting 7,000+ microservices; designed and implemented Kubernetes solution into Azure using Pivotal Kubernetes Service (PKS/TKGI).
- Led the PaaS deployment of a Fiserv presence into Azure; presented at SpringOne 2019 on hybrid-cloud Terraform architecture. Authored Terraform solution for Azure enterprise infrastructure.
- Built a self-service Kubernetes cluster provisioning platform (web form → Git → pipeline → Helm-deployed Prometheus/Grafana stack + expiration cleanup); designed centralized SLI dashboards across 14 PAS foundations using Grafana, AlertManager, and Prometheus.
- Achieved Certified Kubernetes Administrator (CKA-2000-008308-0100) certification; formed and trained the Cloud Platform team using agile methodologies.

Grange Insurance

2006 – 2017 (with gap 2008)

SENIOR SYSTEM ENGINEER – EXCHANGE / ACTIVE DIRECTORY / MESSAGING

- Migrated 3,000 on-premise Exchange 2013 mailboxes to O365; administered 9-server Exchange 2003/2007/2013 environment; architected AD, ADFS, PKI, KMS, PeopleSoft GL integrations, and enterprise FTP platform.
- Built PowerShell and VB automation tooling including a New User Tool, nested AD group reporting, AD photo import pipeline, and SharePoint offboarding automation.

Earlier Roles

1998 – 2008

NETJETS (SR. NETWORK ADMIN) · BMW FINANCIAL (CRM ADMIN) · CARDINAL HEALTH (SR. MESSAGING ENGINEER) · WD PARTNERS (IT ADMIN)

- 25+ years of progressive infrastructure experience spanning messaging, networking, virtualisation, enterprise AD/Exchange, and automation — from first ISP support role (BellSouth.net) through senior engineering leadership.

CERTIFICATIONS

Certified Kubernetes Administrator (CKA)	CKA-2000-008308-0100
Azure DevOps Engineer Expert (AZ-400)	December 2020
Azure Solutions Architect Expert (AZ-303/304)	October 2020
Azure Administrator Associate (AZ-104)	October 2020

EDUCATION & NOTABLE

Franklin University — Computer Science	Columbus, OH
Eagle Scout — Boy Scouts of America	Order of the Arrow
SpringOne 2019 Speaker	Pivotal Platform / Azure
Open Source Contributor	github.com/theautoroboto